FOUR EQUIVALENT CHARACTERIZATIONS OF LOCAL QUASI-FINITENESS FOR A MORPHISM OF ALGEBRAIC STACKS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The characterization of quasi-DM morphisms through locally quasi-finite covers. Invoked inside the counted proof (source lines 5160-5170). Links local quasi-finiteness with quasi-DM-ness in condition (2).
- The criterion for checking a property of a morphism after a flat locally finitely presented cover. Invoked inside the counted proof (source lines 5171-5190). Allows the property to be tested on the covering algebraic space of condition (3).
- The permanence of local quasi-finiteness under composition. Invoked inside the counted proof (source lines 5191-5205). Combines the two morphisms in conditions (3) and (4).

Source reference. Original lemma lemma-characterize-quasi-DM: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 5086–5086. Navigation pointer only: the counted text refers to the Stacks item lemma-characterize-quasi-DM, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-check-finite-type-covering: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3554–3554. Navigation pointer only: the counted text refers to the Stacks item lemma-check-finite-type-covering, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

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Source reference. Original lemma lemma-check-finite-type-precompose: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3586–3586. Navigation pointer only: the counted text refers to the Stacks item lemma-check-finite-type-precompose, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-composition-locally-quasi-finite: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 5037–5037. Navigation pointer only: the counted text refers to the Stacks item lemma-composition-locally-quasi-finite, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-locally-quasi-finite: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4931–4931. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-locally-quasi-finite, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. The following are equivalent*

(1) *f is locally quasi-finite,*
(2) *f is quasi-DM and for any morphism $V \to \mathcal{Y}$ with V an algebraic space and any locally quasi-finite morphism $U \to \mathcal{X} \times_{\mathcal{Y}} V$ where U is an algebraic space the morphism $U \to V$ is locally quasi-finite,*
(3) *for any morphism $V \to \mathcal{Y}$ from an algebraic space V there exists a surjective, flat, locally finitely presented, and locally quasi-finite morphism $U \to \mathcal{X} \times_{\mathcal{Y}} V$ where U is an algebraic space such that $U \to V$ is locally quasi-finite,*
(4) *there exists algebraic spaces U, V, a surjective, flat, and locally of finite presentation morphism $V \to \mathcal{Y}$, and a morphism $U \to \mathcal{X} \times_{\mathcal{Y}} V$ which is surjective, flat, locally of finite presentation, and locally quasi-finite such that $U \to V$ is locally quasi-finite.*

Proof. Assume (1). Then f is quasi-DM by assumption. Let $V \to \mathcal{Y}$ and $U \to \mathcal{X} \times_{\mathcal{Y}} V$ be as in (2). By Lemma lemma-composition-locally-quasi-finite the composition $U \to \mathcal{X} \times_{\mathcal{Y}} V \to V$ is locally quasi-finite. Thus (1) implies (2).

Assume (2). Let $V \to \mathcal{Y}$ be as in (3). By Lemma lemma-characterize-quasi-DM we can find an algebraic space U and a surjective, flat, locally finitely presented, locally quasi-finite morphism $U \to \mathcal{X} \times_{\mathcal{Y}} V$. By (2) the composition $U \to V$ is locally quasi-finite. Thus (2) implies (3).

It is immediate that (3) implies (4).

Assume (4). We will prove (1) holds, which finishes the proof. By Lemma lemma-characterize-quasi-DM we see that f is quasi-DM. To prove that f is locally of finite type it suffices to prove that $g : \mathcal{X} \times_{\mathcal{Y}} V \to V$ is locally of finite type, see Lemma lemma-check-finite-type-covering. Then it suffices to check that g precomposed with $h : U \to \mathcal{X} \times_{\mathcal{Y}} V$ is locally of finite type, see Lemma lemma-check-finite-type-precompose. Since $g \circ h : U \to V$ was assumed to be locally quasi-finite this holds, hence f is locally of finite type. Finally, let k be a field and let $\mathrm{Spec}(k) \to \mathcal{Y}$ be a morphism. Then $V \times_{\mathcal{Y}} \mathrm{Spec}(k)$ is a nonempty algebraic space which is locally of

finite presentation over k. Hence we can find a finite extension k'/k and a morphism $\mathrm{Spec}(k') \to V$ such that

$$\begin{array}{ccc} \mathrm{Spec}(k') & \longrightarrow & V \\ \downarrow & & \downarrow \\ \mathrm{Spec}(k) & \longrightarrow & \mathcal{Y} \end{array}$$

commutes (details omitted). Then $\mathcal{X}_{k'} \to \mathcal{X}_k$ is representable (by schemes), surjective, and finite locally free. In particular $|\mathcal{X}_{k'}| \to |\mathcal{X}_k|$ is surjective and open. Thus it suffices to prove that $|\mathcal{X}_{k'}|$ is discrete. Since

$$U \times_V \mathrm{Spec}(k') = U \times_{\mathcal{X} \times_{\mathcal{Y}} V} \mathcal{X}_{k'}$$

we see that $U \times_V \mathrm{Spec}(k') \to \mathcal{X}_{k'}$ is surjective, flat, and locally of finite presentation (as a base change of $U \to \mathcal{X} \times_{\mathcal{Y}} V$). Hence $|U \times_V \mathrm{Spec}(k')| \to |\mathcal{X}_{k'}|$ is surjective and open. Thus it suffices to show that $|U \times_V \mathrm{Spec}(k')|$ is discrete. This follows from the fact that $U \to V$ is locally quasi-finite (either by our definition above or from the original definition for morphisms of algebraic spaces, via Morphisms of Spaces, Lemma spaces-morphisms-lemma-locally-quasi-finite). □